

Class 5 Restorations

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Class 5 lesions are carious and noncarious defects found in the gingival third of facial and lingual tooth surfaces. As described in chapter 5, Class 5 caries lesions are produced by bacterial plaque attaching to the surface of teeth and producing acids that cause demineralization. A Class 5 lesion resulting from factors other than dental caries is known as a *noncarious cervical lesion* (NCCL). Chapter 13 contains an excellent discussion of the etiology of caries-damaged root surfaces that is applicable to the management and treatment of caries lesions in cervical areas. This chapter focuses mainly on the unique etiology, diagnosis, and restorative treatment of NCCLs (Fig 15-1). For restorative treatment in cervical areas, procedures are similar for caries lesions and noncarious lesions, except that carious dentin is removed in restorative treatment of caries lesions.



Fig 15-1*(a)* The canine has occlusal wear facets but no NCCL. NCCLs in the first premolar are occlusal and gingival to an existing resin composite restoration, suggesting an acid component to the tooth loss. The second premolar has an NCCL. *(b)* V-shaped lesions with a good volume of keratinized attached gingiva apical to the lesions. *(c)* A deep, V-shaped cervical notch in a maxillary canine. *(d)* NCCLs with gingival margins in enamel. *(e)* An isolated NCCL that exposed the pulp in a 70-year-old man. *(f)* Smooth NCCLs involving both enamel and root surfaces. The mandibular molar and second premolar have no keratinized attached gingiva remaining. A gingival cleft has developed on the maxillary second premolar.

Caries Lesions

Tooth color is not a good predictor of root caries damage. A root surface may be discolored and still have a hard, sclerotic surface that would not warrant preparation and placement of a restoration unless the discoloration presented an esthetic problem for the patient. In contrast, some root caries lesions will have the color of healthy tooth structure but will be soft when tested with a dental instrument. Caries-disclosing dyes may be inconsistent in identifying demineralized cementum/dentin on root surfaces. The best correlation to date for clinical detection of caries lesions on root surfaces is the softness of the surface as evaluated with a dental instrument.

Noncarious Cervical Lesions

Incidence

It appears that NCCLs are unique to modern man. In an anthropologic study of the skulls of humans living in the Copper Age and Middle Ages (2050–2080 BC and 1100–1400 AD), no NCCLs were found in 3,927 teeth from 259 individuals.[1]

Today, NCCLs can be found in the teeth of children as well as in adult teeth. Prevalence in various patient and population groups ranges from rare to 89%.[2] Most studies show incidence increasing with age.[2,3] One study[3] involved examination of 1,753 children at 12 years of age and then again at 14 years of age. Cervical or occlusal erosion was present in 56.3% of the children at 12 years of age and 64.1% of the same children at 14 years of age.[3] Another study[4] looked at 1,002 adult inhabitants of Rijeka, Croatia, chosen at random from four dental practices. After the exclusion of restored facial surfaces, 18% of the teeth had evidence of facial tooth wear. In this study, only 5% of subjects younger than 26 years of age were identified with cervical wear. With every 10-year increase in patient age, the lesions became more common. Cervical wear was identified in 48% of subjects older than 65 years of age.[4]

Etiology

The entire etiology of NCCLs has not been determined. While the various possible causes and their degree of involvement may be controversial, there is overwhelming evidence that the cause is multifactorial.[5–10] A review of the literature shows disparate opinions as to the primary cause, with nearly equal evidence to support toothpaste/toothbrush abrasion,[11–16] occlusal forces,[9,13,16–18] and low pH.[19–22] Thus, it is likely that varying combinations of stress factors contribute to the initial lesion and further loss of tooth structure. *Erosion*, the loss of tooth structure from chemical dissolution; *abrasion*, the loss of tooth structure by mechanical or frictional forces; and *abfraction* or *stress corrosion*, the loss of cervical tooth structure due to occlusal forces (Fig 15-2), have all been implicated in the formation of some NCCLs. Patients should be informed of the possible etiologies, implications of the presence of lesions, prevention methods, treatment alternatives, and expected outcomes. Failure to appropriately prevent and treat NCCLs can result in *(1)* progressive loss of tooth structure, *(2)* tooth sensitivity, *(3)* the need for endodontic therapy, *(4)* tooth loss, and *(5)* the occurrence of additional lesions.[23]

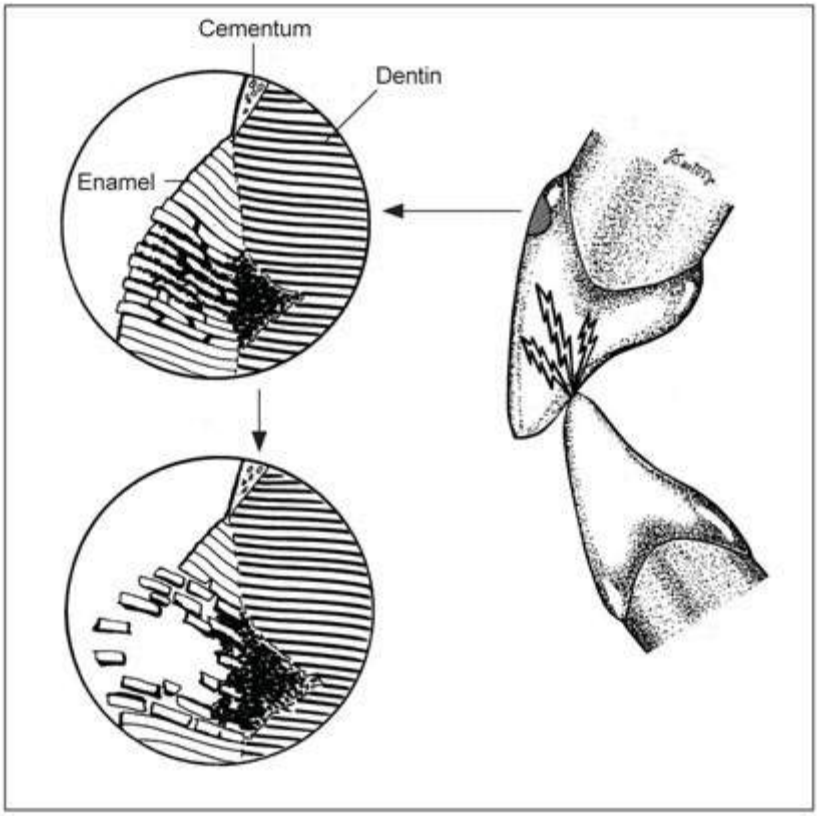


Fig 15-2 The abfraction theory holds that tooth flexure causes loosening of enamel rods, which initiates the cervical lesions.

The ability of ions in saliva to induce remineralization of demineralized tooth structure should be considered an important factor in the inhibition of NCCLs as well as in the inhibition of caries. Caries lesions and noncarious lesions are more common on facial surfaces of teeth than on lingual surfaces. While toothbrush/toothpaste force is a significant etiologic factor, an additional consideration could be differences in the chemistry and character of saliva in lingual and facial areas, which bring about differences in remineralization of tooth structure and the dilution and buffering of acids.[24,25] The effects of xerostomia on oral health are well documented[26–28] and are discussed in chapter 13. However, even in the healthy patient, dehydration from perspiration with physical activity can create impaired salivary flow and inhibit buffering of acids in the oral cavity.[29,30]

In addition to in vivo studies, laboratory studies have contributed to the knowledge base about NCCLs. Finite element analysis has predicted that occlusal loads could cause stress fractures or stress corrosion in the cervical areas of teeth.[31,32] Litonjua et al[33] found that without any acids present, occlusal loading had no effect in creating NCCLs in recently extracted teeth subjected to a toothpaste slurry and tooth brushing at the cervical margins. Brushing with the toothpaste slurry

caused similar cervical wear patterns regardless of whether the tooth was loaded. In contrast, Palamara et al[34] found that enamel dissolution increased significantly when teeth were subjected to cyclic tensile loads while immersed in 1% lactic acid (pH 4.5). The acidic environment and loading made the teeth more susceptible to cervical tissue loss. Volumetric loss was greater in the cervical third than in the middle third and much greater in the mesiobuccal segment (under tension) than in the distobuccal segment (under compression). These findings could offer some explanation for the differences in size, shape, and location of NCCLs.

The morphologic presentation of NCCLs, such as a "wedge" or a "saucer" shape, can perhaps aid the practitioner in discovering their etiology, but the ability to differentiate etiologies based on lesion shape has yet to be adequately supported with evidence.[35] Whitehead et al,[19] studying stress corrosion of enamel at low pH, found that axial loading of extracted premolar teeth in an acid solution resulted in macroscopic and microscopic features similar to those observed in NCCLs in vivo.Changes occurring only in the presence of acids would be consistent with the anthropology samples that did not have NCCLs despite showing evidence of heavy occlusal wear.[1] It appears that liquid acid, frequently found in modern diets, may be necessary to make occlusal loading a factor in the formation of cervical lesions.

Diet

Dietary (extrinsic) acids from foods such as soft drinks, wines, and fruits and fruit juices and gastric (intrinsic) acids that result from gastric reflux, bulimia, and other disorders are both capable of causing tooth dissolution. It is well established in the literature that acids cause dissolution of tooth structure.[23] Soft drinks, which contain citric and/or phosphoric acids, are probably the most frequent source for liquid dietary acids in the US population. Orange juice has also been shown in several studies to cause erosive loss of dentin and enamel.[36,37] When the pH of citric acid and phosphoric acid solutions was adjusted to be similar, citric acid caused more erosion of both enamel and dentin samples than did phosphoric acid.[38]

Toothbrush abrasion

When a tooth is softened by acid dissolution, the effects of any mechanical wear are greatly accelerated. The effect of the acid on enamel and dentin makes the tooth more susceptible to abrasion and attrition (occlusal wear).[20] A study by Attin et al[21] confirmed previous studies[20,22] that showed that eroded enamel is extraordinarily sensitive to mechanical impacts, such as tooth brushing, immediately after

demineralization. The fact that a demineralized tooth surface can be remineralized if not exposed to brushing or other mechanical insult after it has been softened by demineralization is critical to understanding the etiology, prevention, and treatment of NCCLs. Tooth structure can be remineralized after a period of exposure to saliva.[39] These findings are contrary to past, and some current, recommendations by dentists that brushing immediately after a meal or other acid challenge should be routine. On the contrary, tooth structure loss as a result of abrasion can be minimized by delaying brushing by at least 1 hour.[21] In addition, patients with NCCLs should be encouraged to use a minimal amount of dentifrice and light forces when brushing. Anecdotal claims that the majority of toothbrush abrasion lesions appear on the opposite arch from the dominant hand have not been verified with clinical studies.[40]

Occlusion

Of all possible etiologies for NCCLs, occlusal stress forces have received the most attention in recent years.[41] Dentin and enamel have different tensile strengths. With occlusal loading, stress concentration occurs in the cervical area.[18,42] The *abfraction theory*, also known as the *occlusal stress* or *stress corrosion theory*, maintains that tooth flexure in the cervical area results in microfractures of the crystalline structure of the enamel and dentin in that area (see Fig 15-2). The lesion, in theory, would continue to enlarge as the bending and flexing is repeated. Associating wear facets (attrition) with NCCLs is an important component in this theory, and occlusal wear facets may indeed appear in combination with many NCCLs. Studies have found both occlusal wear and NCCLs to occur in the same teeth, but the degree to which this association has been found has varied from 15%[41] to 38%[40] to 95%.[17,43] The lack of consistent findings among studies could be due to variations in the definition of wear facets, differences in the populations studied, or differences in the exclusion criteria applied. While the studies differ on the significance that occlusal forces play in the etiology of NCCLs, no authors claim occlusion to be the solitary etiology for NCCLs.

Treatment approaches

Before any treatment is performed for a patient with one or more NCCLs, a careful examination and determination of possible causes for the lesions should be made. Considering that there is a general consensus in the literature that the etiology is multifactorial, the practitioner's initial approach should not be based on a single assumed cause. The mere presence of occlusal wear facets accompanying NCCLs

does not indicate that the cause is unifactoral (ie, that the lesion is the result of occlusal forces only). Likewise, for a patient with a known history of gastric esophageal reflux disease (GERD), it should not be assumed that the patient's NCCLs are solely the result of a constant intrinsic acid challenge that is causing cervical erosion. While there may be a primary etiology, other contributing factors must also be considered. In all cases of NCCLs, the dentist should obtain a careful health history and provide a thorough dental examination that includes, but is certainly not limited to, an occlusal analysis. The patient's oral hygiene habits should be discussed and evaluated, and a dietary analysis should be considered. Only after determining the probable causes should any treatment commence.[44]

The first goal of any treatment is to remove the causes of the NCCL(s). Once the etiologic factors for a patient are determined, the dentist can help the patient to understand them and to change those that are under his or her control. If a patient is experiencing acute sensitivity associated with one or more lesions, treatment to alleviate the sensitivity should be accomplished. This treatment could involve desensitizing the tooth, restoring the tooth, or possibly performing a periodontal procedure, such as a connective tissue graft, to cover and protect the affected area.[45–52]

Because of the location of Class 5 lesions, access for restorative treatment is often troublesome, moisture control can be exceedingly difficult to obtain and maintain, and soft tissue surgical approaches may be required. Because of the sclerotic nature of the tooth structure in a cervical lesion and the physical properties of restorative materials, long-term retention of the restoration presents a unique challenge. If a decision is made to place a restoration, some lesions can be treated without cavity preparation, and others require preparation to obtain adequate retention of the restoration.

Shallow lesions that lead to thermal sensitivity or sensitivity to touch should first be treated with bonding resins or desensitizing agents. These treatments are low-risk and have reasonable potential for success. Long-term cervical sensitivity studies are very difficult to standardize. Over the course of time, some teeth will become symptom-free without treatment.[53] Some of the self-etching dentin bonding systems show good promise for controlling cervical sensitivity without requiring any habit changes, such as tooth brushing changes and diet modification.[54] Desensitizing toothpaste and fluoride gel formulations are effective for treating cervical sensitivity; these require continuation of use for weeks or months, or longer, to maintain the therapeutic effect.[55] Dental pastes containing 8% arginine and calcium carbonate claim high immediate success for sensitive teeth but appear to require

continuous use to maintain the positive effect.[56] The use of fluoride varnish or fluoride desensitization solutions have proven effective in many situations.[57,58] A recent survey of over 200 private dental practitioners revealed widespread use of all of the treatments mentioned above.[59] But their belief was that fluorides were most successful for treating dentin hypersensitivity and that advice about dieting and tooth brushing were unsuccessful, though they often used this latter approach.

The decision to place a restoration because of a Class 5 lesion is not always easily made. Certainly, if active caries is present, treatment should be initiated to control the active disease and to prevent disease progression. Treatment decisions relating to NCCLs or arrested caries lesions, however, are more difficult. It is generally believed that NCCLs should be treated to protect remaining tooth structure if the amount of tooth structure lost is extensive or progressing, if esthetics is compromised, or to control sensitivity not relieved by less invasive procedures.[60–65] In contrast, a few clinicians believe that all noncarious Class 5 lesions require restorative treatment and describe many reasons for doing so.[62] The Academy of Operative Dentistry has published recommendations for the treatment of NCCLs in an attempt to aid clinicians in the sometimes-difficult treatment decisions and to provide objective guidelines.[23] Their recommendations state that the decision to restore a tooth affected with a NCCL should depend upon the following factors:

- Inability to eliminate or greatly reduce the rate of lesion progression through elimination of etiologic factors
- Esthetic unacceptability of the lesion to the patient
- Significant sensitivity of exposed dentin to cold liquids, food, and air
- Threat to the strength of the tooth and integrity of the coronal-radicular unit because of the lesion's depth

Of course the preferred treatment for a minimal lesion is to eliminate the causes and stop lesion progression. This prevents initiation of the "rerestoration cycle," that is, the repetitious replacement of lost or defective restorations with increasingly larger ones.[66,67]

Periodontal treatment

Periodontal surgery can be used to cover NCCL lesions associated with gingival recession by means of augmenting the gingival tissues in the affected area. While more expensive and more invasive than direct restorations, it is a valid treatment

modality because NCCLs involve exclusively or predominantly the anatomical root of the tooth. Zucchelli et al[68–70] have offered a triage method to determine if a case is favorable for surgical resolution. The authors have shown that the most coronal level of soft tissue advancement achievable with surgical procedures (eg, connective tissue graft, coronally advanced flap) can be determined through a simple procedure. The ideal papilla height is determined as the distance from the apical aspect of the interproximal contact point vertically in an apical direction to a point projected horizontally from the point where the gingival tissues cross the cementoenamel junction (CEJ) at the line angle (Fig 15-3a). This distance is then measured apically from the actual papilla tip and transferred to the facial root surface by the use of an imaginary horizontal line. This process is accomplished independently for both the mesial and distal aspects of the tooth so that two points, one per side, are determined. A smooth, curved line then connects the two points determined from the above procedure and provides the clinician with the maximum coronal root coverage (MRC) (Fig 15-3b). If the MRC is coincident or nearly so with the CEJ, then only root coverage surgery is performed and restorative treatment of the NCCL may not be indicated. If the MRC is apical to the CEJ, then a Class 5 restoration should be accomplished first, with the apical margin of the restoration at or just apical to the MRC. Placing the restoration prior to surgery allows better access for rubber dam and restoration placement, as well as providing a smooth, hard, and convex surface for coronal flap placement. The reader is referred to these papers[68–70] for more detailed information if surgery is under consideration.

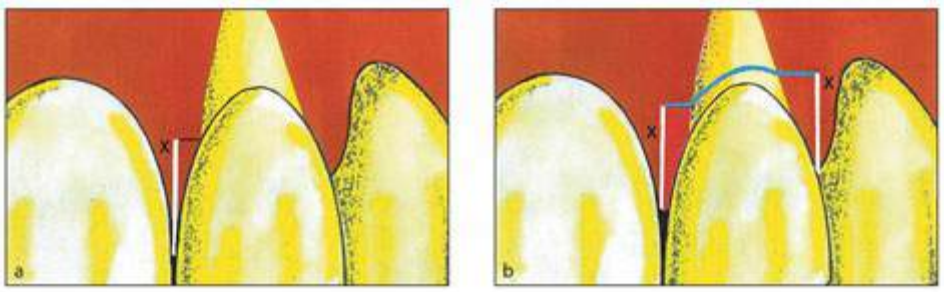


Fig 15-3 When a tooth has gingival recession, the maximum coronal root coverage (MRC) that can be achieved with soft tissue grafting can be determined. *(a)* The ideal papilla height is determined as the distance from the interproximal contact point vertically in an apical direction to a point projected horizontally from the point where the gingival tissues cross the CEJ at the line angle *x*. *(b)* This distance *x* is then measured apically from the actual papilla tip and transferred to the facial root surface by the use of an imaginary horizontal line. This process is accomplished independently for both the mesial and distal aspects of the tooth so that two points, one per side, are determined. A smooth, curved line then connects the two points determined from the above procedure and provides the clinician with the MRC.[68]

Access and Isolation of NCCLs

When cervical lesions occur supragingivally, access to the area for preparation and restoration is often easily obtained. But if the lesion has progressed to or below the free gingival margin, isolation for complete caries removal, tooth preparation, restoration placement, and finishing can be difficult. If a restoration is placed without obtaining complete access to sound tooth structure on all margins, carious tooth structure may remain and the restoration may fail.

One could justify a different restorative goal when the lesion is a noncarious lesion with a margin apical to the gingival crest. Here the restorative goal is to protect the remaining root surface from further damage from toothpaste/toothbrush abrasion without damage to the gingival attachment. A restoration that is smooth but finishes short of the subgingival portion of the lesion might be a good conservative answer. Finishing short is almost certainly superior to placing an overhanging restoration into what the day before was a healthy gingival sulcus. It is well established that a restoration overhang can adversely affect periodontal health.[71] Figure 15-4 shows such an overhang and the associated gingival inflammation.



Fig 15-4*(a)* Gingival inflammation associated with a facial resin composite restoration on the second premolar. *(b)* A retraction cord moves gingiva to reveal the composite resin overhang on a Class 5 restoration.

Nonsurgical retraction

While a rubber dam is the ideal method of field isolation and moisture control for all direct-placement restorations, many Class 5 lesions can be adequately treated using retraction cord (see Fig 15-4b), cotton rolls, and other materials to isolate the lesion and absorb or evacuate moisture. If the lesion extends to or below the gingival margin, a rubber dam is useful to retract the tissue. Often an additional rubber dam–retracting clamp placed directly on the tooth to be restored will provide additional gingival retraction (Fig 15-5). A no. 212SA clamp is effective for this purpose (Fig 15-6a), but modifications to the clamp may be necessary to provide adequate retraction (Figs 15-6b to 15-6d; see also chapter 8).



Fig 15-5 An additional rubber dam clamp is sometimes helpful for retracting both rubber dam and soft tissue.

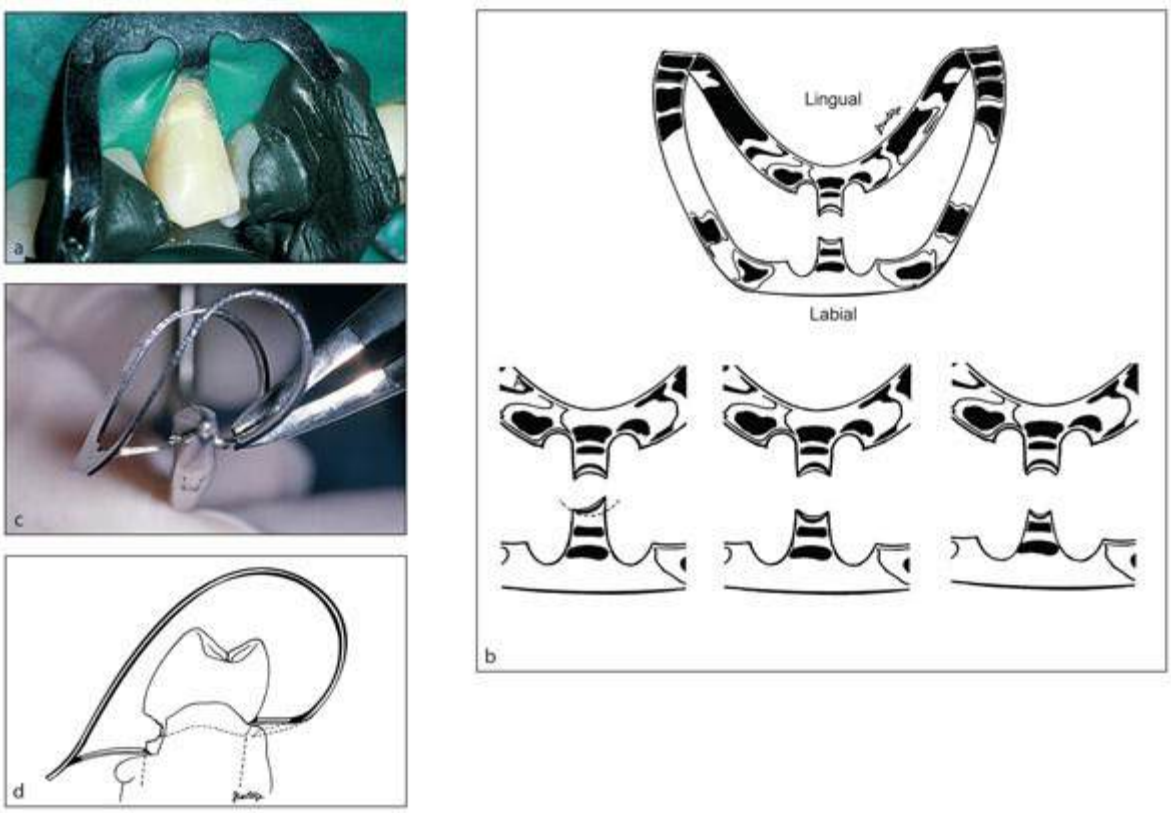


Fig 15-6*(a)* A no. 212SA clamp stabilized with modeling compound. *(b)* Modifications to the no. 212SA clamp are sometimes necessary for proper adaptation to the tooth. *(c)* With two sets of pliers, the lingual jaw of the no. 212SA clamp is bent incisally/occlusally so that it rests on lingual tooth structure, avoiding damage to gingival tissues. *(d)* A modified *(solid line)* and unmodified *(dotted line)* no. 212SA clamp lingual beak.

The clamp must be stabilized to keep it from moving and possibly damaging the restoration or the tooth surface during the operative procedure. Modeling compound is the traditional stabilizing material used (see Figs 15-6a and 8-21b). If lesions in two adjacent teeth are to be treated, modified no. 212SA clamps can be used to provide field isolation (see Figs 15-9d to 15-9h, 8-21d, and 8-21e).

Surgical retraction

Miniflap

As described in chapter 8, the use of miniflaps can often provide sufficient access to subgingival lesions.[72,73] One or two small incisions are made in gingival tissue, beginning at the gingival margin at the mesial and/or distal aspect of the lesion (Fig 15-7; see also Fig 8-48). Each incision is first directed at a right angle to the gingival margin and extended approximately a 1 mm (see Fig 15-7b); the scalpel is then turned so that the remainder of the incision is vertical, approximately parallel with the long axis of the tooth (see Fig 15-7e). It is essential that the entire lesion is exposed, including all demineralized tooth structure. The incision(s) should not be extended past the mucogingival junction. This will allow the small flap of keratinized tissue to be reflected for access, then replaced to the same position after completion of the restoration. Sutures are usually not necessary. If the flap extends past the mucogingival junction, sutures may be required after the restorative procedure has been completed.

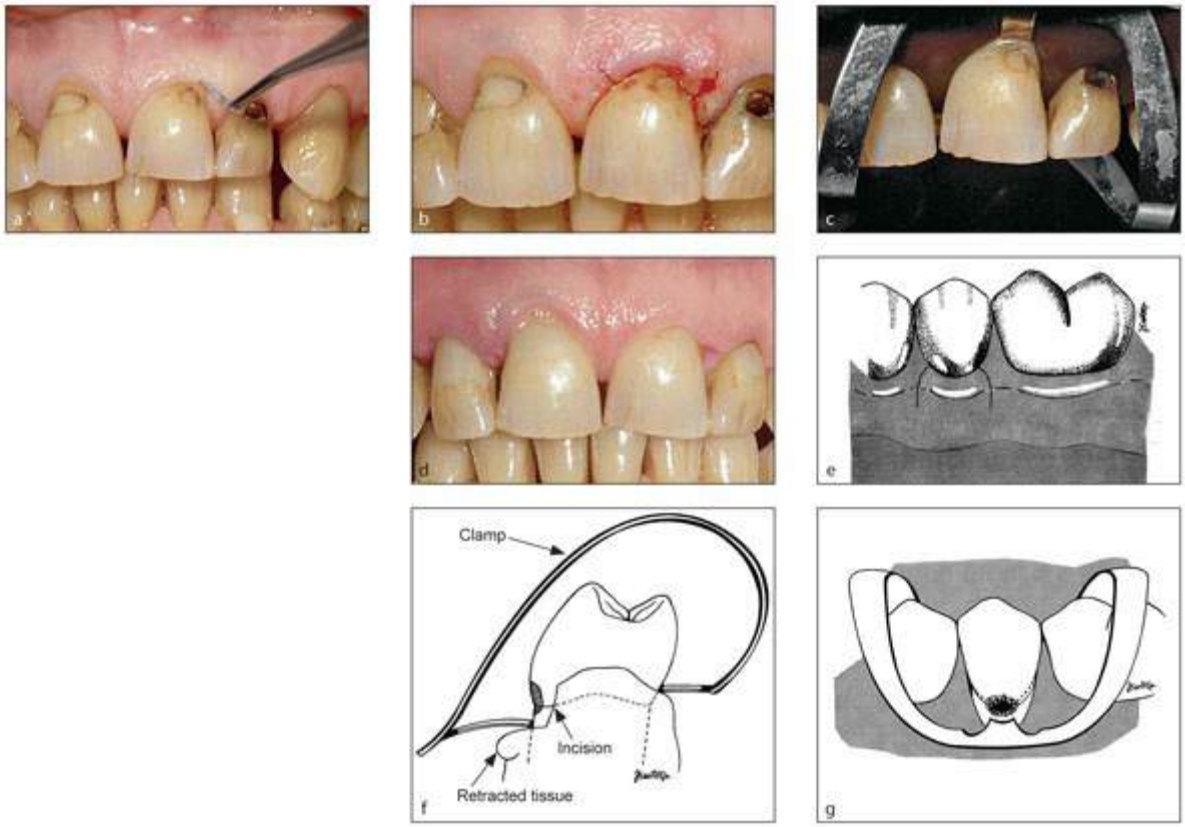


Fig 15-7*(a)* A plastic instrument is used to displace tissue to evaluate the lesion extent and the need for a miniflap. *(b)* Mesial and distal miniflap incisions are made at right angles to the gingival margin. These two incisions are connected with a sulcular incision. *(c)* The lesion is isolated with a rubber dam and a no. 212SA retracting clamp. *(d)* Postoperative appearance of the restoration and

the gingival tissues after 1 week. No sutures were used at the completion of the operative procedure. *(e)* Short vertical incisions are made within the keratinized tissue at right angles to the gingival margin and at the line angles of the tooth. If needed, vertical incisions are made parallel to the long access of the tooth. This allows additional tissue retraction with minimal trauma to the tissue or attachment apparatus. *(f and g)* The no. 212SA retracting clamp and rubber dam are shown in place. This clamp will often need to be stabilized with modeling compound or a similar material.

Gingivoplasty

Gingivoplasty has limited applications because most teeth do not have a sufficiently wide band of keratinized attached gingiva. The width of attached gingiva is determined by subtracting the probing depth from the width of keratinized tissue. If an adequate amount of attached, keratinized gingiva will remain after surgery, a gingivoplasty may be useful in providing access as long as the lesion to be treated will be fully exposed and the biologic width will not be violated[72,73] (Fig 15-8). If gingivoplasty would result in less than 3 mm of attached gingiva adjacent to the restoration, an apically positioned flap or graft procedure may be necessary to preserve or increase the zone of attached gingiva.[74] An inadequate zone of attached gingiva is an absolute contraindication for gingivoplasty.



Fig 15-8*(a)* Carious Class 5 lesions on the maxillary central incisors, requiring surgical access. *(b)* A gingivectomy exposes the full extent of the clinical crowns and provides access to the lesions. *(c)* Maxillary central incisors after placement of restorations. *(d)* Restorations and soft tissue after 2 weeks of healing.

Conventional flap surgery

On occasion, a miniflap would provide insufficient access, so a larger mucoperiosteal flap is required for a cervical restoration.[75–78] If restorations on two adjacent teeth are to be placed simultaneously and both require a surgical procedure for adequate access, a miniflap cannot be used; instead, a mucoperiosteal flap is necessary (Fig 15-9). Surgical crown lengthening with ostectomy may be necessary to provide sufficient access to the lesion and to reestablish dimension for a healthy connective tissue and junctional epithelial attachment[79] (Fig 15-10). A mucoperiosteal flap should be reflected and ostectomy performed as needed. In

some cases, repositioning the envelope flap to its original location will provide the optimal result for esthetics and function. In other cases, the flap will need to be apically positioned. While it is important to remember that the flap will not predictably reattach to the newly placed restoration, there is some evidence to indicate that both soft and hard tissue attachment to glass ionomer or resin-modified glass ionomer (RMGI) does occur. While amalgam has traditionally been used in these applications in posterior teeth, when a restoration is being placed in conjunction with a flap, placement of an all-RMGI restoration or an open sandwich restoration with the apical aspect of the restoration restored with RMGI provides an environment most likely to be conducive to tissue reattachment.[80–85]



Fig 15-9*(a)* Defective Class 5 restorations in the mandibular left canine and first premolar with subgingival margins. *(b)* An envelope flap is reflected, revealing larger defects than were originally evident. *(c)* A rubber dam provides good control of bleeding and moisture but insufficient isolation of the mandibular left canine. *(d)* A modified no. 212SA clamp or a Schultz clamp, stabilized with modeling compound, provides additional retraction and isolation. *(e)* The final preparations. *(f)* Pins were placed in the mandibular left canine for additional retention. *(g)* A custom matrix, stabilized with wedges and compound, provides lateral walls and support for amalgam placement. *(h)* The completed restorations. *(i)* The envelope flap is returned to place and sutured. *(j)* The definitive, polished restorations after 2 weeks of soft tissue healing.

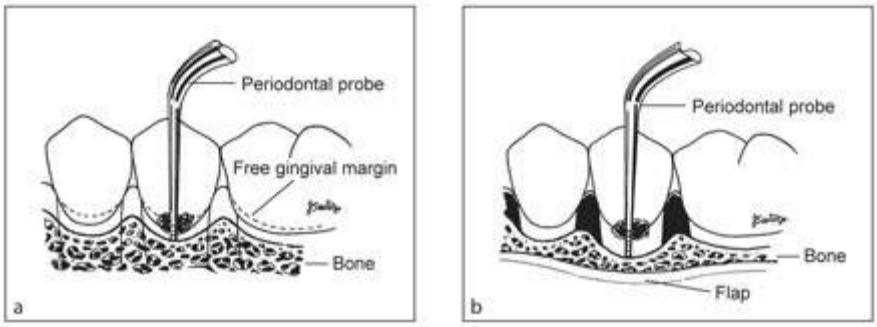


Fig 15-10*(a)* Occasionally, a caries lesion extends to the base of the sulcus, making restoration difficult. Sounding with the periodontal probe shows that the restoration is likely to encroach on the biologic width. Therefore, crown lengthening is indicated. *(b)* A full-thickness mucogingival flap is reflected and bone is removed so that 3 mm of root surface separate the bone and the caries lesion. The restoration can be placed prior to suturing, or restorative procedures can be delayed while the tissue heals.

For posterior teeth, soft tissue rebound is significant following crown lengthening surgery, which frequently results in deep pocket depths. If crown lengthening surgery in a posterior quadrant has been planned, the restorations should be placed 4 to 8 weeks after surgery. Too long of a delay could result in inadequate access to the lesions.[86] However, for anterior teeth, an increased delay between crown lengthening surgery and restoration placement may be indicated, particularly if esthetic reasons dictate that the gingival margins of restorations are located subgingivally. Deas et al[87] showed that the gingival crest height was not fully stable at 6 months.

Timing of surgery

A combined surgical-restorative procedure provides better access to the restorative site than a two-step procedure, in which crown lengthening surgery is accomplished several weeks prior to the restorative procedure. This is because, with the combined procedure, the soft tissue flap is reflected away from the area while the restoration is being placed (Fig 15-11). Moisture and hemorrhage control can best be provided with a well-placed rubber dam. Two interdependent goals are accomplished with a combined surgical procedure: The source of the gingival inflammation is eliminated, and the lost tooth structure is restored. In a broad sense, the tooth and the environment are restored simultaneously.



Fig 15-11 *(a)* A 24-year-old man with numerous Class 5 caries lesions extending subgingivally. *(b)* An envelope flap provides access to the caries lesions. *(c)* After the flap is reflected, a rubber dam is placed. Bleeding is well controlled. *(d)* All carious tooth structure is removed, and provisional restorations are placed for caries control. The flap is apically repositioned. *(e)* One week postoperatively, the tissues are healing and the provisional restorations are coronal to the tissue.

If surgical crown lengthening is to be done prior to restoration of carious root lesions, at least 4 to 8 weeks must elapse after surgery to obtain maturation of the altered periodontium.[74] In esthetic areas, a longer healing period is often required to allow the gingival margin to stabilize before restoration placement.

Cervical Restorations: Materials, Design, and Retention

Once the decision to place a restoration has been made, the dentist must select a restorative material (Box 15-1) and design the cavity preparation. For any Class 5 restoration, the expected extent of the restoration should be determined by the extent of the lesion.

Box 15-1	**Class 5 restorative materials**
Nonesthetic materials	
Amalgam Gold foil (direct, not widely used) Gold inlay (indirect, not widely used)	
Esthetic materials	
Resin composite (with dentin bonding system) Resin composite (with glass-ionomer liner—sandwich technique) Flowable resin composite	

Glass ionomer
RMGI
Compomer
Porcelain inlay (not widely used)

Amalgam

Amalgam preparations will be the same whether the lesion requiring restoration is carious or noncarious. For NCCLs, preparation requires the removal of sound tooth structure to create a box form for amalgam bulk and retention, so the use of adhesive materials is usually preferred.

For the treatment of caries lesions, the preparation should be extended only to remove carious tooth structure and fragile enamel (Fig 15-12). There is no need to create sharp internal line angles or to remove sound dentin for axial depth greater than 1 mm. The cavosurface margins should be as close to 90 degrees as possible. Cavosurface bevels are contraindicated in preparations for amalgam or glass ionomer because of the low edge strength of these materials. With this design, the walls of the Class 5 preparation often diverge because of the curvature of the tooth surface. For nonbonded amalgam restorations, grooves should be placed in the dentin of both the occlusal and gingival walls to help retain the amalgam. In large preparations, pins or other retentive devices may also be beneficial. Amalgam bonding studies of Class 5 restorations have not been accomplished, but clinical studies have reported on restorations subjected to considerably more direct load than Class 5 restorations.[88–93] These studies indicate that bonding is an excellent method to retain amalgam. If the mesial and distal walls are flared so that the amalgam has no lateral walls to confine it for condensation, a custom matrix may be used to facilitate restoration placement and condensation. The simplest method for a facial Class 5 amalgam restoration is to use a hand instrument (Fig 15-13). If the preparation wraps well into the proximal areas, this method may not suffice. Another simple method uses a metal matrix band cut to a length that wraps around the lingual aspect of the tooth and extends slightly facial to the interproximal contacts (see Fig 15-9g). Interproximal wedges are placed to support and stabilize the matrix, and modeling compound may be softened and pressed interproximally if further support is needed.

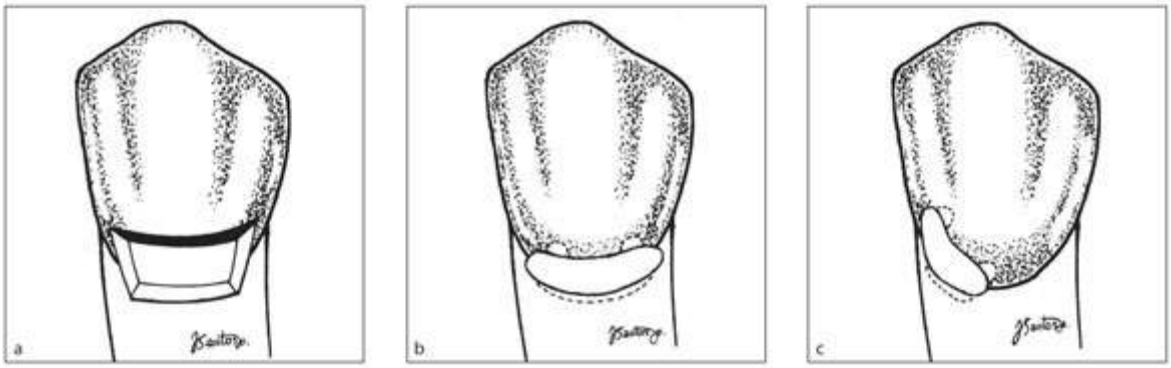


Fig 15-12*(a)* The traditional preparation for a Class 5 direct gold restoration had a trapezoidal outline form that required removal of sound, as well as carious, tooth structure. *(b and c)* The outline form currently recommended for all restorative materials includes removal of carious tooth structure and unsupported enamel. The *dotted lines* indicate retentive undercuts for amalgam restorations. Undercuts are not required when adhesive restorations are placed.



Fig 15-13 A hand instrument may be used as a matrix for Class 5 amalgam restorations.

After the amalgam has been carved to proper contours, a smoother surface may be attained with burnishing and then smoothing with a rubber cup and a fine abrasive paste. Although polishing has been shown to have no effect on marginal ditching,[94] a smooth surface tends to be less plaque retentive.

Bonded tooth-colored restorations

Many techniques and materials have been developed to encourage long-term retention of esthetic materials placed in cervical locations. Stresses due to polymerization shrinkage can cause resin composite to pull away from the tooth-restoration interface, leaving an open margin and pathway for leakage to occur, as has been shown in numerous in vitro studies.[95–98] For moderate-sized to large restorations, incremental resin composite placement is recommended to decrease the effects of polymerization shrinkage[78,99–102] (Fig 15-14).

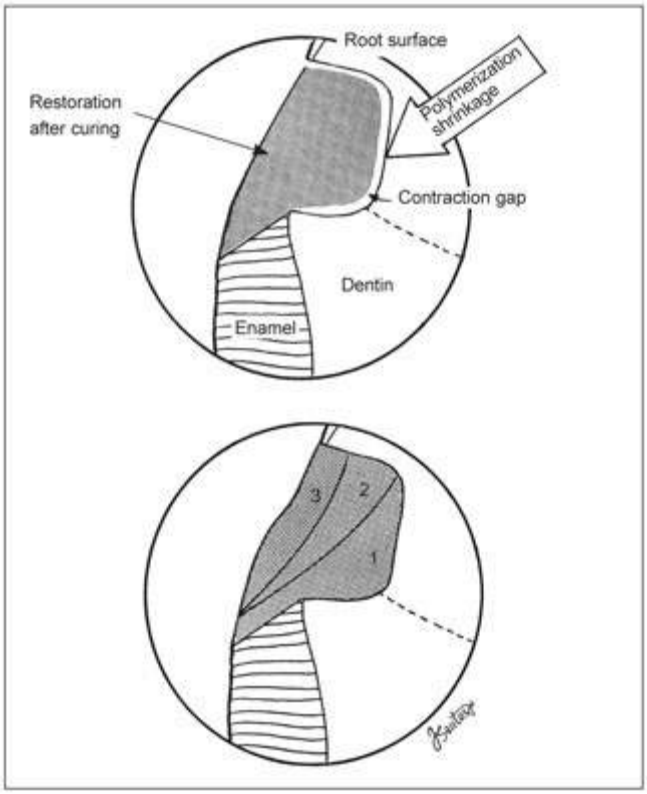


Fig 15-14 If the restoration is large, the resin composite should be placed in at least two increments (1, 2, 3) to compensate for polymerization shrinkage.

If the margins of the restoration will be completely on enamel, the retention of bonded restorations should be predictably successful. Beveling of enamel margins is recommended when it would expose the ends of the enamel rods to provide better etch and bond and/or to improve esthetic blending of the resin composite with the tooth structure.[99,103] Beveling the gingival margin that ends on cementum is not recommended.[104]

An often-overlooked treatment that may improve restoration longevity is occlusal adjustment to reduce eccentric loading of the tooth with the Class 5 restoration.[100] In theory, occlusal adjustment could decrease the dislodging forces placed on the cervical restoration during tooth flexure. To date, there are no controlled clinical trials to verify improved longevity of Class 5 restorations after altering occlusal contacts on the teeth.

Resin composite

The extent and depth of the lesion should determine the outline and depth of the preparation for resin composite, whether the lesion is carious or noncarious. For NCCLs, little or no preparation is required. Retention of a resin composite

restoration is primarily due to the bond, so the bonding system must be used meticulously. Although roughening the surface of a noncarious lesion has been thought to enhance the bond by removing some sclerotic dentin, one clinical trial found no increase in retention when sclerotic lesions were roughened with a bur.[105] Laboratory data have suggested that perhaps an increase in etch time for some etch-and-rinse adhesive systems improves the bond to sclerotic dentin, but depending on the degree of sclerosis, this risks overetching dentin.[106–108] The current self-etching adhesive systems appear to have inferior bonds to sclerotic dentin when matched against dentin bonding systems that require washing away of the etching gel,[109,110] and one clinical study demonstrated the significance of this difference.[111]

Failure to meticulously apply the components of the bonding system could lead to early failure of the restoration. This includes the use of multiple coats of primer if necessary, the application of primer or adhesive with an active scrubbing motion when indicated, the use or non-use of the air syringe for drying, and all other product-specific instructions. For small restorations, the resin composite may be inserted and cured in one increment unless esthetic considerations call for layering to achieve appropriate shading. For restorations that are moderate to large in size, the first increment of resin composite should be placed from about the midpoint of the gingival floor to the incisal or occlusal cavosurface margin and light polymerized. The second increment can then fill the remainder of the preparation.[112] Larger preparations may require more than two increments. Resin composite should be placed in increments no thicker than 2 mm to ensure adequate penetration of light for polymerization.[113] To preserve the cementum or dentin at the gingival margin, careful finishing with a no. 12 or 12B scalpel blade is recommended (Fig 15-15a). Diamond burs, carbide finishing burs, or aluminum oxide disks may be used for contouring (Fig 15-15b). Polishing may be performed with progressively finer-grit disks or abrasive-impregnated rubber points or cups. The highest luster may be achieved with microfilled and nanofilled composites, but most of the current generations of microhybrid and nanohybrid composites also polish well. One study's recommendation[114] to use microfilled composites instead of hybrids in Class 5 restorations because of their lower modulus of elasticity appears to have been refuted. In more recent studies by Browning et al[75] and Van Meerbeek et al,[115] it was concluded that there are no differences in retention rates between microfilled and hybrid resin composites. As with the placement of any resin composite restoration, careful technique is critical to long-term success. Rebonding, as discussed in chapter 11, is recommended for Class 5 restorations.

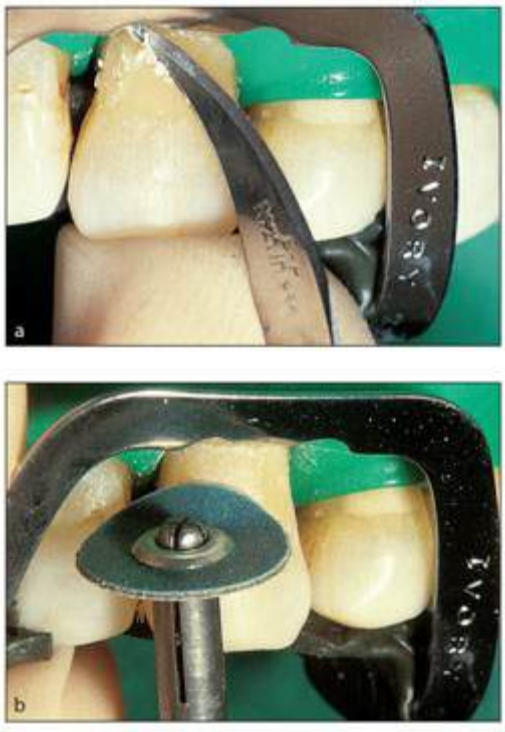


Fig 15-15*(a)* A no. 12 or 12B scalpel blade works well for removing flash. *(b)* Finishing disks may be used to contour and polish Class 5 restorations.

Flowable resin composite

Flowable resin composites have reduced filler particle loading, a lower elastic modulus, significantly higher polymerization shrinkage, a higher coefficient of thermal expansion, and lower fracture toughness relative to traditional resin composites.[116] Flowable resin composites have been recommended for Class 5 restorations with the suggestion that, as the tooth flexes, the less rigid restoration might be able to accommodate the change in cavity shape and therefore be more difficult to dislodge. Clinical trials have proven the theoretical to be incorrect.[75] Stiff restorations have performed as well or better than the flexible resin composites in long-term clinical trials. The use of a flowable resin composite as a liner has not been shown to improve clinical performance.[75,117,118]

Glass ionomer and RMGI

Glass ionomer has been used successfully in Class 5 restorations for many years. One clinical study reported an 80% retention rate of restorations placed without mechanical retention at 10 years.[119] Traditional glass-ionomer materials suffer surface degradation rather rapidly, especially in the presence of acidic foods. Both traditional glass ionomers and RMGIs appear to offer high levels of fluoride release

and excellent recharge capacity. Patients at high risk for caries would probably be well served with either glass-ionomer or RMGI restorations on root surfaces (see chapters 5, 13, and 14).

The preparation for glass-ionomer restorations is similar to that for dental amalgam but without the mechanical retention. Cavosurface bevels are not recommended for the preparation because glass ionomer is a brittle material that requires bulk for strength. A 90-degree butt joint approximately 1 mm deep is a reasonable minimum thickness. After cavity conditioning and placement of RMGI material into the preparation, it is light activated in a manner similar to light curing of resin composite. The use of a clear cervical matrix is optional. Most instructions for use recommend a delay of 2 to 5 minutes before polishing. Franco et al[120] reported that 96.4% of RMGI restorations were retained at 5 years, while only 51.5% of bonded resin composite restorations were retained in the same study. In a more recent study, 94% of Class 5 RMGIs were retained, which exceeded the success of composites (81% to 85%) and far exceeded that of conventional glass ionomers at 67%.[121] A review of Class 5 restorations concluded that glass-ionomer materials (both conventional and resin-modified) showed the highest retention rates.[122]

Glass-ionomer sandwich technique

Because autocured glass-ionomer materials often provide less-than-optimal esthetics[41] (Fig 15-16), some clinicians use the sandwich technique. Glass ionomer is used to replace the missing dentin, reduce leakage,[123–125] improve the potential for tissue attachment for subgingival restorations, and potentially increase retention. A veneer of resin composite is placed to enhance esthetics, increase color stability, improve marginal performance, provide a smoother surface, and increase abrasion resistance[76,126] (Fig 15-17). In one clinical study using the sandwich technique, a 100% retention rate was reported after 3 years.[79] Another clinical trial resulted in a 96% survival rate of sandwich restorations at 5 years, demonstrating the success rates attainable with this type of restoration.[127]



Fig 15-16*(a)* Multiple adjacent Class 5 lesions in the maxillary left canine through the second molar. *(b)* The teeth were restored with conventional glass ionomer, which is opaque and relatively unesthetic.

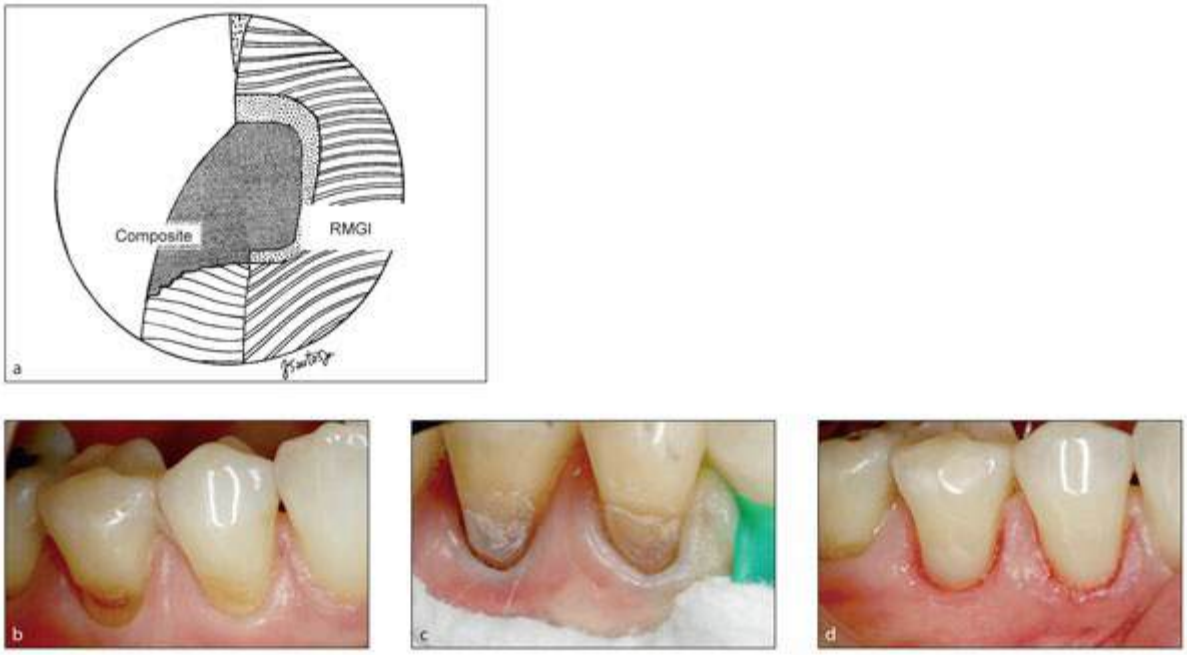


Fig 15-17*(a)* The sandwich technique combines a glass-ionomer base with a veneer of resin composite. *(b)* Class 5 NCCLs on the mandibular right first and second premolars. *(c)* Restorative RMGI (Fuji II LC, GC America) has been placed over all dentin surfaces. The RMGI extends to the gingival dentin margins but is left short of the occlusal enamel margin. The occlusal enamel margin is etched, and a three-step etch-and-rinse adhesive (Optibond FL, Kerr) is placed and cured over the entire lesion, including the RMGI and the etched enamel margin. *(d)* Completed restorations. The RMGI along the axial wall and enamel margin has been veneered with resin composite (Filtek Supreme, 3M ESPE). The apical dentin margin is restored with the RMGI restorative material.

Compomer

Compomer materials are polyacid-modified resin composites (see chapter 14). They are recommended to restore teeth that have carious cervical lesions and NCCLs.[62,63] On the continuum of fluoride-releasing materials, compomer materials fall between resin composites and RMGI materials but are much more like resin

composites. They require the use of a dentin bonding system. Most physical properties of compomers are inferior to those of conventional resin composites; however, compomers have very favorable handling characteristics. Specifically, their lack of "stickiness" has brought them ready acceptance in the marketplace. However, the marginal integrity of compomers has been worse than that of resin composites in long-term clinical trials.[128–141]

Dentinal Sensitivity

Dentinal sensitivity, a problem reported to affect anywhere from 8% to 74% of the population,[142–144] is often associated with gingival recession and NCCLs.[79] A survey of new patients found that, of 780 patients, 32% reported sensitivity, but only 5% had sought professional help.[145] Tooth sensitivity was the chief complaint in 60% of the referrals to a periodontal specialty clinic.[146] Sensitivity is caused by exposure of dentinal tubules that communicate between the pulp and the oral cavity; the degree of sensitivity is influenced by the number and size of the open tubules.[3,147] The *hydrodynamic theory*[148] is the most widely accepted explanation of dentinal sensitivity. Changes in the direction of fluid movement within open dentinal tubules are perceived as pain by mechanoreceptors near the pulp. Tactile, thermal, or osmotic stimuli can induce changes in fluid flow and elicit a pain response.

In a clinical study, caries-free teeth that were planned for extraction were stimulated to determine if exposed dentin was hypersensitive. Microscopic analysis of the teeth found that the teeth that were hypersensitive had eight times more open or patent tubules per unit area, and the tubule diameters in the sensitive teeth were twice as wide.[149]

Treatment or prevention of hypersensitivity is usually accomplished by the use of some method to occlude the open dentinal tubules.[59,150–152] Dentin adhesives provide at least short-term relief.[153–155] Stannous fluorides have also been used with positive results.[57,156] A study found that 10% strontium chloride solution, 2% sodium fluoride solution, and 40% for-malin solution significantly reduced dentin hypersensitivity, whereas a 5% solution of potassium nitrate did not.[157] A potassium chloride–containing chewing gum was found to be effective in the treatment of hypersensitive teeth.[158]

A recent systematic review concluded that there was insufficient clinical evidence to recommend oxalates for treating dentin hypersensitivity.[159] No treatment to date has been effective for every patient every time. The variety of treatments available

would suggest that, by using proper diagnostic and clinical skills, the dentist should be able to help most patients who have sensitive teeth.

Summary

The preponderance of evidence supports the conclusion that the loss of cervical enamel, cementum, and dentin is due to a combination of many factors. Contributing factors include the mechanical action of the toothpaste and toothbrush, intake of acidic foods, reflux of gastric acids into the oral cavity, bacteria-produced acids, inadequate buffering ability of saliva, inadequate salivary flow rate, and mechanical abrasion. While not completely understood, occlusal forces inducing stresses in the cervical areas of teeth may play a role in the development of cervical lesions. When the etiologic factors have been reduced or eliminated and all other conservative management of cervical lesions has been attempted, there may still be a need to restore areas where there has been significant cervical tooth loss. The decision to restore should be tempered by the certainty that restorations are not permanent.

References

2. Aubry M, Mafart B, Donat B, Brau JJ. Brief communication: Study of noncarious cervical tooth lesions in samples of prehistoric, historic, and modern populations from the South of France. Am J Phys Anthro-pol 2003;121:10–14.
2. Li P, Li J, Zhang Q. Investigations on the incidence of dental wedge-shaped defects and abrasion in 501 elderly population and their correlation [in Chinese]. Zhonghua Kou Qiang Yi Xue Za Zhi 1998;33:227–229.
3. Dugmore CR, Rock WP. The progression of tooth erosion in a cohort of adolescents of mixed ethnicity. Int J Paediatr Dent 2003;13:295–303.
4. Borcic J, Anic I, Urek MM, Ferreri S. The prevalence of non-carious cervical lesions in permanent dentition. J Oral Rehabil 2004;31:117–123.
5. Bader JD, McClure F, Scurria MS, Shugars DA, Heymann HO. Case-control study of non-carious cervical lesions. Community Dent Oral Epidemiol 1996;24:286–291.
6. Grippo J, Simring M, Coleman T. Abfraction, abrasion, biocorrosion, and the enigma of noncarious cervical lesions: A 20-year perspective. J Esthet Restor Dent 2012;24:10–25.
7. Wood I, Jawad Z, Paisley C, Brunton P. Non-carious cervical tooth surface loss: A literature review. J Dent 2008;36:759–766.
8. Osborne-Smith KL, Burke FJ, Wilson NH. The aetiology of the non-carious cervical lesion. Int Dent J 1999;49:139–143.
9. Ceruti P, Menicucci G, Mariani GD, Pittoni D, Gassino G. Non carious cervical lesions. A review. Minerva Stomatol 2006;55:43–57.
10. Pecie R, Krejci I, Garcia-Godoy F, Bortolotto T. Noncarious cervical lesions—A clinical concept based on the literature review. Part 1: Prevention. Am J Dent 2011;24:49–56.
11. Litonjua LA, Andreana S, Bush PJ, Tobias TS, Cohen RE. Wedged cervical lesions produced by toothbrushing. Am J Dent 2004;17:237–240.
12. Bartlett DW, Shah P. A critical review of non-carious cervical (wear) lesions and the role of abfraction,

erosion, and abrasion. J Dent Res 2006;85:306–312.
13. Smith WA, Marchan S, Rafeek RN. The prevalence and severity of non-carious cervical lesions in a group of patients attending a university hospital in Trinidad. J Oral Rehabil 2008;35:128–134.
14. Michael JA, Townsend GC, Greenwood LF, Kaidonis JA. Abfraction: Separating fact from fiction. Aust Dent J 2009;54:2–8.
15. Pikdöken L, Akca E, Gürbüzer B, Aydil B, Taşdelen B. Cervical wear and occlusal wear from a periodontal perspective. J Oral Rehabil 2011;38:95–100.
16. Jiang H, Du MQ, Huang W, Peng B, Bian Z, Tai BJ. The prevalence of and risk factors for non-carious cervical lesions in adults in Hubei Province, China. Community Dent Health 2011;28:22–28.
17. Miller N, Penaud J, Ambrosini P, Bisson-Boutelliez C, Briançon S. Analysis of etiologic factors and periodontal conditions involved with 309 abfractions. J Clin Periodontol 2003;30:828–832.
18. Palamara D, Palamara JE, Tyas MJ, Messer HH. Strain patterns in cervical enamel of teeth subjected to occlusal loading. Dent Mater 2000;16:412–419.
19. Whitehead SA, Wilson NH, Watts DC. Development of noncarious cervical notch lesion in vitro. J Esthet Dent 1999;11:332–337.
20. Attin T, Koidl U, Buchalla W, Schaller HG, Kielbassa AM, Hellwig E. Correlation of microhardness and wear in differently eroded bovine dental enamel. Arch Oral Biol 1997;42:243–250.
21. Attin T, Buchalla W, Gollner M, Hellwig E. Use of variable remineralization periods to improve the abrasion resistance of previously eroded enamel. Caries Res 2000;34:48–52.
22. Davis WB, Winter PJ. The effect of abrasion on enamel and dentine and exposure to dietary acid. Br Dent J 1980;148:253–256.
23. American Academy of Operative Dentistry. Non-carious cervical lesions. Recommendations for clinical practice. Oper Dent 2003;28:109–113.
24. Dawes C. Physiological factors affecting salivary flow rate, oral sugar clearance, and the sensation of dry mouth in man. J Dent Res 1987;66:648–653.
25. Young WG. Anthropology, tooth wear, and occlusion ab origine. J Dent Res 1998;77:1860–1863.
26. Astroth JD. Caring for the elderly adult. How to prevent, manage root surface caries. Dent Teamwork 1996;9:15–19.
27. Atkinson JC, Wu AJ. Salivary gland dysfunction: Causes, symptoms, treatment. J Am Dent Assoc 1994;125:409–416.
28. Haveman CW, Redding SW. Dental management and treatment of xerostomic patients. Tex Dent J 1998;115(6):43–56.
29. Young WG, Khan F. Sites of dental erosion are saliva-dependent. J Oral Rehabil 2002;29:35–43.
30. Khan F, Young WG, Daley TJ. Dental erosion and bruxism. A tooth wear analysis from south east Queensland. Aust Dent J 1998;43:117–127.
31. Rees JS. The effect of variation in occlusal loading on the development of abfraction lesions: A finite element study. J Oral Rehabil 2002;29:188–193.
32. Geramy A, Sharafoddin F. Abfraction: 3D analysis by means of the finite element method. Quintessence Int 2003;34:526–533.
33. Litonjua LA, Bush PJ, Andreana S, Tobias TS, Cohen RE. Effects of occlusal load on cervical lesions. J Oral Rehabil 2004;31:225–232.
34. Palamara D, Palamara JE, Tyas MJ, Pintado M, Messer HH. Effect of stress on acid dissolution of enamel. Dent Mater 2001;17:109–115.
35. Maseki T, Tanaka H. Symmetry of non-carious cervical lesions in canines and premolars. Gerodontology 2006;23:183–186.
36. Hughes JA, Jandt KD, Baker N, et al. Further modification to soft drinks to minimise erosion. A study in situ. Caries Res 2002;36:70–74.
37. Absi EG, Addy M, Adams D. Dentine hypersensitivity—The effect of toothbrushing and dietary compounds on dentine in vitro: An SEM study. J Oral Rehabil 1992;19:101–110.
38. West NX, Hughes JA, Addy M. The effect of pH on the erosion of dentine and enamel by dietary acids in

vitro. J Oral Rehabil 2001;28:860–864.
39. Jaeggi T, Lussi A. Toothbrush abrasion of erosively altered enamel after intraoral exposure to saliva: An in situ study. Caries Res 1999;33:455–461.
40. Oginni AO, Olusile AO, Udoye CI. Non-carious cervical lesions in a Nigerian population: Abrasion or abfraction? Int Dent J 2003;53:275–279.
41. Piotrowski BT, Gillette WB, Hancock EB. Examining the prevalence and characteristics of abfractionlike cervical lesions in a population of U.S. veterans. J Am Dent Assoc 2001;132:1694–1701.
42. Kuroe T, Itoh H, Caputo AA, Konuma M. Biomechanics of cervical tooth structure lesions and their restoration. Quintessence Int 2000;31:267–274.
43. Mayhew RB, Jessee SA, Martin RE. Association of occlusal, periodontal, and dietary factors with the presence of non-carious cervical dental lesions. Am J Dent 1998;11:29–32.
44. LittleStar ML, Summitt JB. Non-carious cervical lesions: An evidenced-based approach to their diagnosis. Tex Dent J 2003;120:972–980.
45. Majzoub Z, Landi L, Grusovin MG, Cordioli G. Histology of connective tissue graft. A case report. J Periodontol 2001;72:1607–1615.
46. Saletta D, Pini Prato G, Pagliaro U, Baldi C, Mauri M, Nieri M. Coronally advanced flap procedure: Is the interdental papilla a prognostic factor for root coverage? J Periodontol 2001;72:760–766.
47. Ito K, Murai S. Adjacent gingival recession treated with expanded polytetrafluoroethylene membranes: A report of 2 cases. J Periodontol 1996;67:443–450.
48. Cordioli G, Mortarino C, Chierico A, Grusovin MG, Majzoub Z. Comparison of 2 techniques of subepithelial connective tissue graft in the treatment of gingival recessions. J Periodontol 2001;72:1470–1476.
49. Novaes AB Jr, Grisi DC, Molina GO, Souza SL, Taba M Jr, Grisi MF. Comparative 6-month clinical study of a subepithelial connective tissue graft and acellular dermal matrix graft for the treatment of gingival recession. J Periodontol 2001;72:1477–1484.
50. Rosetti EP, Marcantonio RA, Rossa C Jr, Chaves ES, Goissis G, Marcantonio E Jr. Treatment of gingival recession: Comparative study between subepithelial connective tissue graft and guided tissue regeneration. J Periodontol 2000;71:1441–1447.
51. Tözüm TF, Dini FM. Treatment of adjacent gingival recessions with sub-epithelial connective tissue grafts and the modified tunnel technique. Quintessence Int 2003;34:7–13.
52. Harris RJ. Connective tissue grafts combined with either double pedicle grafts or coronally positioned pedicle grafts: Results of 266 consecutively treated defects in 200 patients. Int J Periodontics Restorative Dent 2002;22:463–471.
53. Morris MF, Davis RD, Richardson BW. Clinical efficacy of two dentin desensitizing agents. Am J Dent 1999;12:72–76.
54. Baysan A, Lynch E. Treatment of cervical sensitivity with a root sealant. Am J Dent 2003;16:135–138.
55. Schiff T, Zhang YP, DeVizio W, et al. A randomized clinical trial of the desensitizing efficacy of three dentifrices. Compend Contin Educ Dent Suppl 2000;27:4–10.
56. Hamlin D, Mateo LR, Dibart S, Delgado E, Zhang YP. Comparative efficacy of two treatment regimens combining in-office and at-home programs for dentin hypersensitivity relief: A 24-week clinical study. Am J Dent 2012;25:146–152.
57. Thrash WJ, Dodds MW, Jones DL. The effect of stannous fluoride on dentinal hypersensitivity. Int Dent J 1994;44(suppl 1):107–118.
58. Thrash WJ, Jones DL, Dodds WJ. Effect of a fluoride solution on dentinal hypersensitivity. Am J Dent 1992;5:299–302.
59. Cunha-Cruz J, Wataha JC, Zhou L, et al. Treating dentin hypersensitivity: Therapeutic choices made by dentists of the northwest PRECEDENT network. J Am Dent Assoc 2010;141:1097–1105.
60. Baum LG, Phillips RW, Lund MR. Textbook of Operative Dentistry, ed 3. Philadelphia: Saunders, 1995:210–212,262.
61. Crispin BJ. Contemporary Esthetic Dentistry: Practice Fundamentals. Chicago: Quintessence, 1994:112.
62. Grippo JO. Noncarious cervical lesions: The decision to ignore or restore. J Esthet Dent 1992;4(suppl):55–64.

63. Levitch LC, Bader JD, Shugars DA, Heymann HO. Non-carious cervical lesions. J Dent 1994;22:195–207.
64. Powell LV, Gordon GE, Johnson GH. Sensitivity restored of Class V abrasion/erosion lesions. J Am Dent Assoc 1990;121:694–696.
65. Sturdevant CM, Roberson TM, Heymann HO, Sturdevant JR. The Art and Science of Operative Dentistry, ed 3. St Louis: Mosby, 1995:486–499.
66. Brantley CF, Bader JD, Shugars DA, Nesbit SP. Does the cycle of reresto-ration lead to larger restorations? J Am Dent Assoc 1995;126:1407–1413.
67. Fontana M, González-Cabezas C. Secondary caries and restoration replacement: An unresolved problem. Compend Contin Educ Dent 2000;21:15–18,21–24,26.
68. Zucchelli G. Testori T, De Sanctis M. Clinical and anatomical factors limiting treatment outcomes of gingival recession: A new method to predetermine the line of root coverage. J Periodontol 2006;77:714–721.
69. Zucchelli G, Gori G, Mele M, et al. Non-carious cervical lesions associated with gingival recessions: A decision-making process. J Periodontol 2011;82:1713–1724.
70. Zucchelli G, Mele M, Stefanini M, et al. Predetermination of root coverage. J Periodontol 2010;81:1019–1026.
71. Parsell DE, Streckfus CF, Stewart BM, Buchanan WT. The effect of amalgam overhangs on alveolar bone height as a function of patient age and overhang width. Oper Dent 1998;23:94–99.
72. Reagan SE. Periodontal access techniques for restorative dentistry. Gen Dent 1989;37:117–121.
73. Starr CB. Management of periodontal tissues for restorative dentistry. J Esthet Dent 1991;3:195–208.
74. Maynard JG Jr, Wilson RD. Physiologic dimensions of the periodontium significant to the restorative dentist. J Periodontol 1979;50:170–174.
75. Browning WD, Brackett WW, Gilpatrick RO. Two-year clinical comparison of a microfilled and a hybrid resin-based composite in non-carious Class V lesions. Oper Dent 2000;25:46–50.
76. Suzuki M, Jordan RE. Glass ionomer-composite sandwich technique. J Am Dent Assoc 1990;120:55–57.
77. Vandewalle KS, Vigil G. Guidelines for the restoration of Class V lesions. Gen Dent 1997;45:254–260.
78. McCoy RB, Anderson MH, Lepe X, Johnson GH. Clinical success of Class V composite resin restorations without mechanical retention. J Am Dent Assoc 1998;129:593–599.
79. Powell LV, Johnson GH, Gordon GE. Factors associated with clinical success of cervical abrasion/erosion restorations. Oper Dent 1995;20:7–13.
80. Kolhatkar S, Haque SA, Winkler JR, Bhola M. Root coverage in an HIV-positive individual: Combined use of a lateral sliding flap and resin-modified glass ionomer for the management of an isolated severe recession defect. J Periodontol 2010;81:632–640.
81. Santamaria MP, Suaid FF, Casati MZ, Nociti FH, Sallum AW, Sallum EA. Coronally positioned flap plus resin-modified glass ionomer restoration for the treatment of gingival recession associated with non-carious cervical lesions: A randomized controlled clinical trial. J Periodontol 2008;79:621–628.
82. Alkan A, Keskiner I, Yuzbasioglu E. Connective tissue grafting on resin ionomer in localized gingival recession. J Periodontol 2006;77:1446–1451.
83. Heithersay GS. Invasive cervical resorption following trauma. Aust Endod J 1999;25:79–85.
84. White C Jr, Certosimo AJ. Combining reconstructive and regenerative therapies. J Am Dent Assoc 1997;128:625–629.
85. Dragoo MR. Resin-ionomer and hybrid-ionomer cements: Part II, human clinical and histologic wound healing responses in specific periodontal lesions. Int J Periodontics Restorative Dent 1997;17:75–87.
86. Pontoriero R, Carnevale G. Surgical crown lengthening: A 12-month clinical wound healing study. J Periodontol 2001;72:841–848.
87. Deas DE, Moritz AJ, McDonnell HT, Powell CA, Mealey BL. Osseous surgery for crown lengthening: A 6-month clinical study. J Periodontol 2004;75:1288–1294.
88. Belcher MA, Stewart GP. Two-year clinical evaluation of an amalgam adhesive. J Am Dent Assoc 1997;128:309–314.
89. Summitt JB, Burgess JO, Berry TG, Robbins JW, Osborne JW, Haveman CW. Six-year clinical evaluation of bonded and pin-retained complex amalgam restorations. Oper Dent 2004;29:261–268.
90. Evans DB, Neme AM. Shear bond strength of composite resin and amalgam adhesive systems to dentin. Am

J Dent 1999;12:19–25.
91. Görücü J, Tiritoglu M, Ozgünaltay G. Effects of preparation designs and adhesive systems on retention of Class II amalgam restorations. J Prosthet Dent 1997;78:250–254.
92. Oliveira JP, Cochran MA, Moore BK. Influence of bonded amalgam restorations on the fracture strength of teeth. Oper Dent 1996;21:110–115.
93. Staninec M, Marshall GW, Lowe A, Ruzickova T. Clinical research on bonded amalgam restorations. Gen Dent 1997;45:356–362.
94. Mayhew RB, Schmeltzer LD, Pierson WP. Effect of polishing on the marginal integrity of high-copper amalgam. Oper Dent 1986;11:8–13.
95. Carvalho RM, Pereira JC, Yoshiyama M, Pashley DH. A review of polymerization contraction: The influence of stress development versus stress relief. Oper Dent 1996;21:17–24.
96. Choi KK, Condon JR, Ferracane JL. The effects of adhesive thickness on polymerization contraction stress of composite. J Dent Res 2000;79:812–817.
97. Ferracane JL, Mitchem JC. Relationship between composite contraction stress and leakage in Class V cavities. Am J Dent 2003;16:239–243.
98. Calheiros FC, Sadek FT, Braga RR, Cardoso PE. Polymerization contraction stress of low-shrinkage composites and its correlation with microleakage in Class V restorations. J Dent 2004;32:407–412.
99. Albers H. Tooth-Colored Restoratives, ed 8. Santa Rosa, CA: Alto Books, 1996:8c1–8c6.
100. Leinfelder KF. Restoration of abfracted lesions. Compendium 1994; 15:1396–1400.
101. Lutz E, Krejci I, Oldenburg TR. Elimination of polymerization stresses at the margins of posterior composite resin restorations: A new restorative technique. Quintessence Int 1986;17:777–784.
102. Tjan AH, Bergh BH, Lidner C. Effect of various incremental techniques on the marginal adaptation of Class II composite resin restorations. J Prosthet Dent 1992;67:62–66.
103. Hilton TJ, Ferracane JL. Cavity preparation factors and microleakage of Class III composite restorations filled at intraoral temperatures. Am J Dent 1998;11:123–130.
104. Owens BM, Halter TK, Brown DM. Microleakage of tooth colored restorations with a beveled gingival margin. Quintessence Int 1998;29:356–361.
105. Van Dijken JW. Clinical evaluation of three adhesive systems in Class V non-carious lesions. Dent Mater 2000;16:285–291.
106. Lopes GC, Baratieri CM, Baratieri LN, Monteiro S Jr, Cardoso Vieira LC. Bonding to cervical sclerotic dentin: Effect of acid etching time. J Adhes Dent 2004;6:19–23.
107. Lopes GC, Vieira LC, Monteiro S Jr, Caldeira de Andrada MA, Baratieri CM. Dentin bonding: Effect of degree of mineralization and acid etching time. Oper Dent 2003;28:429–439.
108. Lopes GC, Vieira LC, Arújo E, Bruggmann T, Zucco J, Oliveira G. Effect of dentin age and acid etching time on dentin bonding. J Adhes Dent 2011;13:139–145.
109. Kwong SM, Cheung GS, Kei LH, et al. Micro-tensile bond strengths to sclerotic dentin using a self-etching and a total-etching technique. Dent Mater 2002;18:359–369.
110. Tay FR, Kwong SM, Itthagarun A, et al. Bonding of a self-etching primer to non-carious cervical sclerotic dentin: Interfacial ultrastructure and microtensile bond strength evaluation. J Adhes Dent 2000;2:9–28.
111. Ritter AV, Heymann HO, Swift EJ Jr, Sturdevant JR, Wilder AD Jr. Clinical evaluation of an all-in-one adhesive in non-carious cervical lesions with different degrees of dentin sclerosis. Oper Dent 2008;33:370–378.
112. Crim GA. Microleakage of three resin placement techniques. Am J Dent 1991;4:69–72.
113. Jensen ME, Chan DC. Polymerization shrinkage and microleakage. In: Vanherle G, Smith DC (eds). Posterior Resin Composite Dental Restorative Materials. Utrecht, The Netherlands: Szulc, 1985:243–262.
114. Heymann HO, Sturdevant JR, Bayne S, Wilder AD, Sluder TB, Brunson WD. Examining tooth flexure effects on cervical restorations: A two-year clinical study. J Am Dent Assoc 1991;122:41–47.
115. Van Meerbeek B, Kanumilli PV, De Munck J, Van Landuyt K, Lambrechts P, Peumans M. A randomized, controlled trial evaluating the three-year clinical effectiveness of two etch & rinse adhesives in cervical lesions. Oper Dent 2004;29:376–385.
116. ADA Council on Scientific Affairs. Restorative materials. ADA Professional Product Review 2010;5(2):1–

16.
117. Ernst CP, Buhtz C, Rissing C, Willershausen B. Clinical performance of resin composite restorations after 2 years. Compend Contin Educ Dent 2002;23:711–714,716–717,720.
118. Tyas MJ, Burrow MF. Three-year clinical evaluation of One-Step in non-carious cervical lesions. Am J Dent 2002;15:309–311.
119. Matis BA, Cochran M, Carlson T. Longevity of glass-ionomer restorative materials: Results of a 10-year evaluation. Quintessence Int 1996;27: 373–382.
120. Franco EB, Benetti AR, Ishikiriama SK, et al. 5-year clinical performance of resin composite versus resin modified glass ionomer restorative system in non-carious cervical lesions. Oper Dent 2006;31:403–408.
121. Stewardson DA, Thornley P, Bigg T, et al. The survival of Class V restorations in general dental practice. Part 2, early failure. Br Dent J 2011;210:E19.
122. Heintze SD, Roulet JF. Glass ionomer derivates have better retention rates in cervical restorations compared to self-etching adhesive systems. J Evid Based Dent Pract 2010;10:18–20.
123. Brackett WW, Gilpatrick RO, Browning WD, Gregory PN. Two-year clinical performance of a resin-modified glass-ionomer restorative material. Oper Dent 1999;24:9–13.
124. Abdalla AI, Alhadainy HA. Clinical evaluation of hybrid ionomer restoratives in Class V abrasion lesions: Two-year results. Quintessence Int 1997;28:255–258.
125. Folwaczny M, Loher C, Mehl A, Kunzelmann KH, Hickel R. Class V lesions restored with four different tooth-colored materials—3-year results. Clin Oral Investig 2001;5:31–39.
126. Frazier KB, Rueggeberg FA, Mettenburg DJ. Comparison of wear-resistance of Class V restorative materials. J Esthet Dent 1998;10:309–314.
127. Tyas MJ, Burrow MF. Clinical evaluation of a resin-modified glass ionomer adhesive system: Results at five years. Oper Dent 2002;27:438–441.
128. Van Dijken JW. 3-year clinical evaluation of a compomer, a resin-modified glass ionomer and a resin composite in Class III restorations. Am J Dent 1996;9:195–198.
129. Huth KC, Manhart J, Selbertinger A, et al. 4-year clinical performance and survival analysis of Class I and II compomer restorations in permanent teeth. Am J Dent 2004;17:51–55.
130. Hse KM, Wei SH. Clinical evaluation of compomer in primary teeth: 1-year results. J Am Dent Assoc 1997;128:1088–1096.
131. Andersson-Wenckert IE, Folkesson UH, Van Dijken JW. Durability of a polyacid-modified composite resin (compomer) in primary molars. A multicenter study. Acta Odontol Scand 1997;55:255–260.
132. Roeters JJ, Frankenmolen F, Burgersdijk RC, Peters TC. Clinical evaluation of Dyract in primary molars: 3-year results. Am J Dent 1998;11:143–148.
133. Prati C, Chersoni S, Cretti L, Montanari G. Retention and marginal adaptation of a compomer placed in non-stress-bearing areas used with the total-etch technique: A 3-year retrospective study. Clin Oral Investig 1998;2:168–173.
134. Papagiannoulis L, Kakaboura A, Pantaleon F, Kavvadia K. Clinical evaluation of a polyacid-modified resin composite (compomer) in Class II restorations of primary teeth: A two-year follow-up study. Pediatr Dent 1999;21:231–234.
135. Di Lenarda R, Cadenaro M, De Stefano Dorigo E. Cervical compomer restorations: The role of cavity etching in a 48-month clinical evaluation. Oper Dent 2000;25:382–387.
136. Van Dijken JW. Durability of new restorative materials in Class III cavities. J Adhes Dent 2001;3:65–70.
137. Abdalla AI, Mahallawy SE, Davidson CL. Clinical and SEM evaluations of three compomer systems in Class V carious lesions. J Oral Rehabil 2002;29:714–719.
138. Folwaczny M, Mehl A, Kunzelmann KH, Hickel R. Clinical performance of a resin-modified glass-ionomer and a compomer in restoring non-carious cervical lesions. 5-year results. Am J Dent 2001;14:153–156.
139. Wucher M, Grobler SR, Senekal PJ. A 3-year clinical evaluation of a compomer, a composite and a compomer/composite (sandwich) in Class II restorations. Am J Dent 2002;15:274–278.
140. Huth KC, Manhard J, Hickel R, Kunzelmann KH. Three-year clinical performance of a compomer in stress-bearing restorations in permanent posterior teeth. Am J Dent 2003;16:255–259.

141. Qvist V, Laurberg L, Poulsen A, Teglers PT. Class II restorations in primary teeth: 7-year study on three resin-modified glass ionomer cements and a compomer. Eur J Oral Sci 2004;112:188–196.
142. Krauser JT. Hypersensitive teeth. Part 1: Etiology. J Prosthet Dent 1986; 56:153–156.
143. Gillam DG, Tang JY, Mordan NJ, Newman HN. The effects of a novel Bioglass dentifrice on dentine sensitivity: A scanning electron microscopy investigation. J Oral Rehabil 2002;29:305–313.
144. Bartold PM. Dentinal hypersensitivity: A review. Aust Dent J 2006; 51:212–218.
145. Liu HC, Lan WH, Hsieh CC. Prevalence and distribution of cervical dentin hypersensitivity in a population in Taipei, Taiwan. J Endod 1998; 24:45–47.
146. Taani SD, Awartani F. Clinical evaluation of cervical dentin sensitivity (CDS) in patients attending general dental clinics (GDC) and periodontal specialty clinics (PSC). J Clin Periodontol 2002;29:118–122.
147. Cuenin MF, Schiedt MS, O'Neal RB, et al. An in vivo study of dentin sensitivity: The relation of dentin sensitivity and the patency of dentin tubules. J Periodontol 1991;62:668–678.
148. Brännström M, Lindén LA, Aström A. The hydrodynamics of the dental tubule and of pulp fluid. A discussion of its significance in relation to dentinal sensitivity. Caries Res 1967;1:310–317.
149. Absi EG, Addy M, Adams D. Dentine hypersensitivity. A study of the patency of dentinal tubules in sensitive and non-sensitive cervical dentine. J Clin Periodontol 1987;14:280–284.
150. Hodosh M, Hodosh SH, Hodosh AJ. About dentinal hypersensitivity. Compendium 1994;15:658,660,662.
151. Pashley DH, Muzzin K. Clinical management of dentin sensitivity. In: Clark JW (ed). Clinical Dentistry. Philadelphia: Lippincott, 1990:1–7.
152. Trowbridge HO, Silver DR. A review of current approaches to in-office management of tooth hypersensitivity. Dent Clin North Am 1990;34: 561–581.
153. Dondi dall'Orologio G, Malferrari S. Desensitizing effects of Gluma and Gluma 2000 on hypersensitive dentin. Am J Dent 1993;6:283–286.
154. Watanabe T, Sano M, Itoh K, Wakumoto S. The effects of primers on the sensitivity of dentin. Dent Mater 1991;7:148–150.
155. Swift EJ Jr, May KN Jr, Mitchell S. Clinical evaluation of Prime & Bond 2.1 for treating cervical dentin hypersensitivity. Am J Dent 2001;14:13–16.
156. Tarbet WJ, Silverman G, Fratarcangelo PA, Kanapka JA. Home treatment for dentinal hypersensitivity: A comparative study. J Am Dent Assoc 1982;105:227–230.
157. Kishore A, Mehrotra KK, Saimbi CS. Effectiveness of desensitizing agents. J Endod 2002;28:34–35.
158. Krahwinkel T, Theiss P, Willershausen B. Clinical effectiveness of a potassium chloride containing chewing gum in the treatment of hypersensitive teeth. Eur J Med Res 2001;6:483–487.
159. Cunha-Cruz J, Stout JR, Heaton LJ, Wataha JC; Northwest PRECEDENT. Dentin hypersensitivity and oxalates: A systematic review. J Dent Res 2011;90:304–310.